

Foundational Requirements for Artificial General Intelligence: A Falsifiable Framework Based on Signal Prediction

Matej Šprogar

University of Maribor, Faculty of Electrical Engineering and Computer Science, Maribor, Slovenia
matej.sprogar@um.si

Abstract—Grounded in the premise that high-level intelligence can emerge from low-level signal processing, we advance a hypothesis regarding low-level requirements necessary for artificial general intelligence. The proposed requirements characterise core properties of systems that learn through prediction over spatially and temporally structured signals with initially unknown semantic content. They include a selection of basic principles observed in cognitive neuroscience, from learning from an uninformed state to real-time liveness.

To enable empirical testing and hypothesis rejection, we introduce an operational testbed composed of transparent and reusable tests, one per requirement. To date, no non-intelligent system has been identified or reported as successfully passing the testbed. Pending such a counterexample, the testbed serves as a candidate empirical milestone toward general intelligence. The reference implementation of the testbed is publicly available.

Index Terms—artificial general intelligence, predictive processing, temporal prediction, pattern recognition, symbol grounding

I. INTRODUCTION

Intelligence remains one of the deepest unresolved questions in science, as we still lack both a precise definition of intelligence and an understanding of its fundamental components [1], [2], [3]. Assuming there is such a thing as general intelligence, one major obstacle to creating artificial general intelligence (AGI) is the limited understanding of how biological brains operate. At the same time, modern artificial intelligence (AI) systems achieve impressive performance across a wide range of tasks [4]. However, such performance is often achieved without a principled account of understanding and may not reflect broadly general capabilities [5], [6], [7].

Until we discover the secret behind general intelligence, our models can only aim to simulate the observable behaviour it produces. As a result, modern AI conflates specialised competence with general intelligence [8], [9], [10]. Systems optimised for a narrow class of tasks may perform extremely well within that domain while remaining brittle or ineffective elsewhere.

The existing definitions and descriptions of intelligent behaviour are useless for AGI development because they are either vague, self-referential, or non-general. However valuable modern AI is, maybe AGI is not achievable by targeting a high-level behaviour on a predefined task. Maybe we should focus on the low-level behaviours necessary for the emergence

of general intelligent behaviour. This perspective is motivated in part by observations from biological neural systems, whose operation is often described in terms of continuous streams of sensory signals rather than discrete symbolic¹ representations [12], [13]. Working directly with low-level signals offers a way to avoid assuming pre-existing semantic structure and allows us to propose testable, operational requirements for the emergence of intelligent behaviour.

In this paper, we identify requirements hypothesised to be necessary but not sufficient for general intelligence and formalise them as architecture-agnostic criteria. Although each requirement can be satisfied individually by specialised systems, complexity arises from their interaction; consequently, the challenge lies in satisfying them simultaneously. We propose eleven requirements as a minimal set hypothesised to capture the essential conditions for general intelligence.

Because intelligence lacks a precise definition, the hypothesis cannot be directly verified and is therefore formulated in a falsifiable manner. To this end, we introduce the Artificial General Intelligence Testbed (AGITB), which evaluates whether a system under test satisfies all proposed requirements. Successfully passing the testbed does not, by itself, establish the presence of general intelligence. However, the existence of a non-intelligent system that satisfies all requirements would falsify the hypothesis. Until such a counterexample is identified, passing the AGITB may be regarded as a meaningful milestone toward AGI, whereas failing it indicates that a system remains limited to narrow or specialised capabilities.

The originality of this work lies not in any individual component considered in isolation, but in the synthesis of minimal, low-level, architecture-independent behavioural requirements. Formalising these requirements enabled the construction of a transparent, reusable testbed for the empirical evaluation of experimental AGI systems, thereby addressing a longstanding methodological gap in the field.

II. RELATIONSHIP TO PRIOR RESEARCH

The present work intersects several established research traditions, including predictive processing and predictive coding

¹Symbolic representations introduce an additional complication known as the symbol grounding problem, in which symbolic tokens require grounding in perception or environmental interaction to acquire intrinsic semantic meaning [11].

in computational neuroscience [14], [12], continual and sequence learning [15], [16], and formal theories of intelligence [17], [18]. However, to our knowledge, no existing framework combines the specific set of objectives pursued here within a single operational, falsifiable system.

Conceptually, the framework is most closely aligned with predictive-processing accounts of cognition, in which intelligent behaviour emerges through continual prediction over temporally structured input streams [19], [12]. Related ideas also appear in sequence-memory approaches such as Hierarchical Temporal Memory [16], where prediction and temporal state evolution are treated as central computational primitives. Existing approaches of this kind, however, are primarily architectural or neurocomputational proposals rather than attempts to derive architecture-independent necessary conditions for general intelligence.

At a more abstract level, the work also relates to formal theories of intelligence, including universal intelligence and AIXI-style formulations [17], [18], which similarly seek principled characterisations of intelligence from minimal assumptions. Unlike the present work, however, these approaches do not operationalise their assumptions as architecture-independent, falsifiable criteria for empirical evaluation.

Our approach further differs from contemporary benchmark-oriented paradigms for evaluating intelligence, including task-generalisation benchmarks such as ARC [20]. Modern machine-learning evaluations are typically constructed around finite test distributions, with hidden or periodically refreshed datasets intended to limit optimisation toward the evaluation itself. By contrast, the proposed testbed does not rely on benchmark secrecy; instead, it introduces transparent, comprehensible tasks. Success depends on satisfying the underlying behavioural constraints across combinatorially large input spaces rather than on reproducing finite hidden test distributions.

III. BACKGROUND

Contemporary artificial systems do not yet robustly exhibit behaviour indistinguishable from that of humans. One possible research direction towards that goal involves closer alignment with the computational principles observed in the human cortex, motivating interest in neuromorphic approaches such as spiking neural networks, which explicitly incorporate time-sensitive, event-driven computation [21], [22]. While such architectures do not, in themselves, constitute general intelligence, they highlight alternative design dimensions that remain relatively underexplored in conventional artificial neural networks.

In alignment with this biologically grounded perspective, we depart from symbolic, high-level behaviour and instead operate at the level of raw signals. While Turing [1] was right to suggest that communication could serve as a basis for evaluating machine intelligence, natural language remains problematic as a test medium: it conveys human knowledge through symbols whose meanings are not intrinsically grounded in machines, as argued by Stevan Harnad [11]. As a result, symbolic evaluation can obscure whether a system has acquired grounded

representations or merely exploits statistical regularities in linguistic form. Although the symbol-grounding problem is an old philosophical issue, it has regained prominence across cognitive science, neuroscience, and machine learning [8], [9], [23].

A. Biological grounding and non-symbolic evaluation

To minimise assumptions about semantic grounding, we focus on systems that operate directly on the lowest-level binary signals, each represented as a structure of bits processed in parallel. A bit corresponds directly to the state of a measurable signal channel (such as a pixel, an audio band, or actuator feedback) and, as the smallest unit of information, does not in itself impose higher-level semantic structure. Binary signals also provide a simple abstraction of neural spiking events in biological systems.

This perspective is consistent with the functioning of the cerebral cortex, where cognition arises from the processing of temporally structured sensory spike trains rather than disembodied symbols. The “brain simulator” reply to John Searle’s Chinese Room argument [24] suggests that a system reproducing the causal dynamics of cortical processing could, in principle, instantiate the same cognitive mechanisms [25]. The cortex represents at least one plausible route toward general intelligence.

The proposed AGI requirements are particularly well-suited for NeuroAI models and align with the principles of the embodied Turing Test [26], in which cognitive understanding is hypothesised to emerge from the integration of continuous sensory streams. The progression from raw-signal prediction to higher-level abstraction mirrors that from early perceptrons to contemporary large-scale models.

B. Integrated requirements for intelligence

A starting point for the AGI requirements is the *ladder to human-comparable intelligence* [27], a framework that organises cognitive capabilities into progressively more demanding levels. Its first three rungs align naturally with the framework proposed here. However, the ladder’s individual requirements can be met by specialised systems that lack general intelligence, which creates a risk of false positives during evaluation. Minimising such false positives, therefore, requires an all-or-nothing criterion: a system must satisfy *all* requirements simultaneously rather than meeting them individually. In addition, the higher rungs of the ladder remain insufficiently specified for operational evaluation.

Consequently, failure on any single requirement is sufficient to rule out AGI, whereas satisfying all of them does not establish it. This complementarity allows individual requirements to remain simple in scope, while their joint satisfaction captures the complexity of general intelligence.

C. Learning through prediction

An intelligent system learns to recognise the structure of input signals originating from the external world. The learning process cannot rely solely on externally defined performance

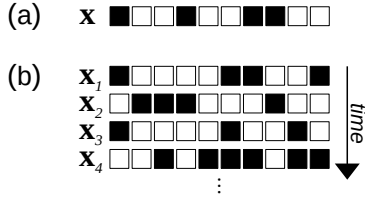


Fig. 1. (a) Example of a 10-bit input $\mathbf{x}$ with four bits set. (b) Example of an input sequence.

metrics, such as accuracy or mean-squared error, because predefined quantitative metrics struggle to capture the richness and variability of real-world structure.

This limitation is reflected in many existing approaches to evaluating AGI, which remain constrained by metrics that inadequately capture the breadth of general intelligence. Contemporary systems already surpass humans on many standardised tasks—for example, playing chess—illustrating how narrowly defined benchmarks can be optimised without necessarily yielding broadly general capabilities. Metric-based evaluation can therefore encourage systems engineered to maximise specific scores rather than those exhibiting general intelligence.

Achieving generality, therefore, requires a learning process driven by an internally defined objective that does not rely on externally supplied semantics. One such objective is the ability to predict the future sensory signals the system will experience. Prediction at the level of sensory signals provides a foundation for higher-level expectations about the external world to emerge without requiring predefined semantic representations. Biological intelligence is often characterised as extracting structure from sensory data in order to anticipate future states [16], [19].

Accordingly, we require a generally intelligent system to predict future inputs from past observations. Prediction alone does not constitute intelligence, but the ability to anticipate future sensory inputs is widely considered a fundamental capability underlying perception, learning, and action [28], [29].

D. Spatial and temporal structure of input signals

In the operational setting considered here, the requirements are specified in terms of temporal sequences comprising randomly structured inputs. Because input signals are defined independently of any external meaning, they need not resemble real-world sensory data. This design limits the usefulness of pretraining and increases confidence that any observed learning arises from the input stream itself.

Each input represents a snapshot of multiple binary signals at a single time step (Figure 1a). Spatial organisation within each input encodes local structure, whereas semantic richness arises from the temporal evolution of the input sequence (Figure 1b). The interaction between spatial and temporal dimensions gives rise to structured patterns that are challenging to adapt to.

An AGI system must learn from a stream of inputs. Upon receiving an input at time step t , it is required to issue a

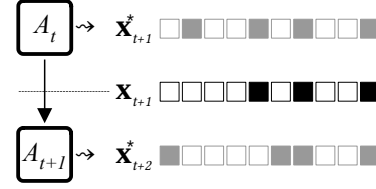


Fig. 2. Iterative adaptation in discrete time. Following input $\mathbf{x}_t$, the system A_t issues the prediction $\mathbf{x}_{t+1}^*$. Upon observing the realised input $\mathbf{x}_{t+1}$, it updates in response to the error in the second bit and produces the next one-step-ahead prediction $\mathbf{x}_{t+2}^*$.

prediction $\mathbf{x}_{t+1}^*$ for the subsequent input, as shown in Figure 2. A transition from configuration A_t to A_{t+1} is triggered by the arrival of the actual input ($\mathbf{x}_{t+1}$). The central challenge is not simply extrapolation but discerning the underlying causes or regularities that produce the observed input stream and using that understanding to make accurate future predictions.

IV. THE HYPOTHESIS

We hypothesise that any system capable of human-level general intelligence must satisfy at least the eleven requirements defined later. These requirements characterise fundamental properties of adaptive systems that learn from temporally structured inputs without relying on externally supplied semantic knowledge. The selection of these requirements is guided by observations of biological information processing and by the need to support autonomous learning from sensory signals.

The hypothesis is empirically falsifiable. It would be rejected by the existence of a demonstrably non-intelligent system that fulfils all requirements. The proposed requirements should therefore not be interpreted as a definition of intelligence.

In what follows, learning denotes the adaptive process through which a system becomes able to correctly anticipate future inputs based on its input history. This definition serves only to characterise prediction success as an operational capability, not an evaluation metric for intelligence. The term *model* denotes a functioning instance of a system type that updates its internal configuration and produces a prediction of a future input based on its input history. Independent models share the same operating principles but are otherwise isolated from one another.

V. PRELIMINARIES

Let $\mathbf{x} \in X = \{0, 1\}^L$ with $L = 10$ denote a ten-bit input vector, and let $\chi = (\mathbf{x}_t)_{t=1}^N$, $\mathbf{x}_t \in X$ denote an input sequence with N elements. For any input vector $\mathbf{x} \in X$, we write $\mathbf{x}[i]$ for its i -th bit.

Let $\mathcal{M}$ denote the set of all reachable² configurations of a system under AGITB evaluation. Individual system instance is denoted by $A \in \mathcal{M}$, with $B \in \mathcal{M}$ representing an independent instance of the same type. When temporal indexing is required, we write A_t for the configuration obtained after t update steps

²Starting from the initial configuration under some input history.

along its learning trajectory, with A_0 representing the initial configuration at the onset of learning.

For brevity, we refer to an instance at a given internal configuration simply as a model.

Definition 1 (Prediction). A model's prediction is given by the operator $\rightsquigarrow \subseteq \mathcal{M} \times X$, where

$$A_t \rightsquigarrow \mathbf{x}_{t+1}^*$$

denotes that the model configuration $A_t \in \mathcal{M}$ predicts the next input to be $\mathbf{x}_{t+1}^* \in X$.

Definition 2 (Model update). Model update is given by the transition relation $\mapsto \subseteq \mathcal{M} \times X \times \mathcal{M}$, where

$$A_t \xrightarrow{\mathbf{x}_{t+1}} A_{t+1}$$

denotes the atomic transition of the model configuration $A_t \in \mathcal{M}$ upon receiving the input $\mathbf{x}_{t+1} \in X$.

Operationally, prediction and update are tightly coupled: processing the input $\mathbf{x}_{t+1}$ both updates the model and produces a prediction of the subsequent input $\mathbf{x}_{t+2}$. This allows us to write the combined transition as

$$A_t(\mathbf{x}_{t+1}) \rightarrow (A_{t+1}, \mathbf{x}_{t+2}^*),$$

which subsumes both model advancement and prediction.

Definition 3 (Model trajectory). For any model configuration $A \in \mathcal{M}$ and any input sequence $\chi = (\mathbf{x}_i)_{i=1}^k$, we write

$$A \xrightarrow{\chi} A^\chi$$

to denote the transitions to the configuration obtained by sequentially updating the model with all inputs in χ , i.e.,

$$A^\chi = A^{(\mathbf{x}_1, \dots, \mathbf{x}_k)} := (\dots ((A \xrightarrow{\mathbf{x}_1} \xrightarrow{\mathbf{x}_2}) \dots) \xrightarrow{\mathbf{x}_k}).$$

Accordingly, A_t^χ denotes the configuration obtained by applying χ to A_t , that is, $A_t^\chi := (A_t)^\chi$. A model may be exposed to multiple input sequences in succession; for two sequences χ_1 and χ_2 , their concatenation is written $\chi_1\chi_2$, and the resulting configuration satisfies

$$A \xrightarrow{\chi_1} A^{\chi_1} \xrightarrow{\chi_2} A^{\chi_1\chi_2}.$$

Definition 4 (Autoregressive generation). Autoregressive generation is given by the relation $\Rightarrow \subseteq \mathcal{M} \times X^{\leq \mathbb{N}}$, where

$$A_t \Rightarrow \chi^*$$

denotes that the model configuration A_t generates the (finite or infinite) sequence $\chi^* = (\mathbf{x}_{t+1}^*, \mathbf{x}_{t+2}^*, \dots)$ by recursively feeding each predicted input back into the model. In particular, for all $k \geq 0$,

$$A_{t+k} \rightsquigarrow \mathbf{x}_{t+k+1}^*, \quad A_{t+k} \xrightarrow{\mathbf{x}_{t+k+1}^*} A_{t+k+1}.$$

Definition 5 (Learning). The model A is said to *learn* a sequence χ , written $A \triangleright \chi$, if there exists a finite number of learning steps after which A can autoregressively reproduce a single instance of χ , i.e.

$$A \triangleright \chi : \Longleftrightarrow \exists n \in \mathbb{N}_{\geq 1} : A^{\chi^n} \Rightarrow \chi.$$

The particular choice of n is immaterial when only the existence of a post-learning configuration is required.

Definition 6 (Post-learning configuration). If $A \triangleright \chi$, we write A_χ for any configuration A^{χ^n} ($n \geq 1$) satisfying $A^{\chi^n} \Rightarrow \chi$.

Definition 7 (Learning time). The *learning time in atomic steps* is defined as

$$\tau_A(\chi) := \begin{cases} |\chi| \cdot \min\{n \geq 1 \mid A^{\chi^n} \Rightarrow \chi\}, & \text{if } A \triangleright \chi, \\ \infty, & \text{else.} \end{cases}$$

Learning time measures the first occurrence of accurate prediction and does not presuppose permanent retention. If the model never learns to predict the sequence, its learning time is infinite.

Definition 8 (Match score). For two sequences $\alpha = (\mathbf{a}_1, \dots, \mathbf{a}_m)$ and $\beta = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ in X^m , define the (unnormalised) bitwise match score

$$S(\alpha, \beta) := \sum_{j=1}^m \sum_{i=1}^L \mathbf{1}\{\mathbf{a}_j[i] = \mathbf{b}_j[i]\}.$$

VI. THE REQUIREMENTS

Requirement 1 (Uninformed start). *All models begin learning from the identical initial configuration $\diamond$.*

$$\forall A : A_0 = \diamond.$$

The initial model is assumed to contain no environment-specific knowledge beyond the architectural biases necessary for learning. All knowledge relevant to input must be acquired solely through subsequent exposure to data. This assumption is commonly regarded as a necessary condition for universality, defined as the ability to adapt to arbitrary environments.

We assume that general-purpose learning systems, including biological brains, do not begin with an innate understanding of external inputs but instead acquire meaning through interaction with their environment. Each system must construct semantic content from raw sensory data rather than rely on pre-encoded knowledge.³

Requirement 2 (Determinism). *Model evolution is deterministic with respect to input.*

$$\forall A, \forall \mathbf{x} \in X : \exists! A' \text{ such that } A \xrightarrow{\mathbf{x}} A'.$$

Biological neurons operate in a functionally deterministic manner, ensuring stability and consistency in brain function. Although minor stochastic effects may occur, they do not undermine the rule-governed nature of neural processing. By analogy, we assume that input history uniquely determines the model configuration. A model is therefore fully determined by its input history.

Determinism at the level of neural signal processing is necessary for stable, reproducible brain function. In contrast, the apparent unpredictability of cognition stems from the

³Although reflexes may be genetically specified, they do not constitute genuine understanding. They are rather evolutionary features of the subcortical "old" brain and not prerequisites for intelligence [16, p. 66].

system's complexity rather than from genuine indeterminacy [30].

Requirement 3 (Trace). *Each input leaves a permanent internal trace.*

$$\forall t \neq s \geq 0 : A_t \neq A_s.$$

A model's internal configurations evolve without recurrence: configurations never repeat, trajectories contain no cycles, and every input leaves a permanent internal trace.

Human brains satisfy an effective version of this requirement at the lowest physical level. Each sensory or internal event induces irreversible microstate changes that are never exactly revisited. Through complex internal interactions, this persistent history contributes to the apparent unpredictability of decision-making. Although information may appear discarded, compressed, or behaviourally inaccessible at the cognitive level, we posit that internal model configurations continue to develop in a distinct manner over time, consistent with the formation of permanent internal traces.

Requirement 4 (Time). *Model evolution depends on input order.*

$$\forall A, \forall \chi_1 \neq \chi_2 \in X^+ : A^{\chi_1 \chi_2} \neq A^{\chi_2 \chi_1}.$$

Model evolution depends intrinsically on input order: for any system, changing the input order necessarily results in a novel configuration. This enforces a strict temporal asymmetry in learning dynamics and rules out commutative or order-invariant update mechanisms. Sensitivity to temporal structure in this strong sense is regarded as a defining property of intelligent systems.

Definition 9 (Admissible sequences). The set of *admissible* sequences is defined as

$$\Phi := \left\{ (\mathbf{x}_1, \dots, \mathbf{x}_k) \in X^k \mid 1 \leq k \leq k_{\max} \right. \\ \left. \wedge \forall i \in \{1, \dots, k\} : \mathbf{x}_i \cdot \mathbf{x}_{((i \bmod k)+1)} = 0 \right\}.$$

Requirement 5 (Absolute refractory period). *A model can learn a cyclic sequence only if successive inputs are pairwise non-overlapping.*

$$\forall \chi : A_0 \triangleright \chi \implies \chi \in \Phi.$$

Biological intelligence relies on discrete spiking events for communication and learning, and individual neurons cannot fire again immediately after activation. The absolute refractory-period constraint defines admissible sequences but does not imply that all such sequences are learnable.

Absolute refractory periods impose a minimum separation between spikes, creating temporal structure that supports temporally sensitive learning processes [22]. All temporal sequences that satisfy biologically plausible refractory constraints are therefore admissible, independent of any semantic interpretation of the signals.

Definition 10 (Learnable sequences). The set of *learnable* sequences is the subset

$$\Psi := \{ \phi \in \Phi \mid A_0 \triangleright \phi \}.$$

Requirement 6 (Inevitable saturation).

(a) *A model cannot learn everything there is to learn.*

$$\forall k \in \mathbb{N}_{>1}, \forall \psi_1, \dots, \psi_k \in \Psi, \exists \psi_{k+1} \in \Psi : \\ \neg \left((((A_0)_{\psi_1})_{\psi_2}) \dots \psi_k \triangleright \psi_{k+1} \right).$$

(b) *All admissible length-2 sequences are universally learnable.*

$$\forall A, \forall \phi \in \Phi \cap X^2 : A \triangleright \phi.$$

Learning systems with finite representational and adaptive capacity necessarily face limits on the extent to which they can learn sequences of length greater than two, whereas all admissible sequences of length two are universally learnable.

Length-two sequences occupy a special position, as they are the shortest sequences that encode temporal relationships, while a single input carries no temporal information. They define transitions that are the irreducible units of temporal prediction and thus necessary for temporal learning.

A related distinction is observed in biological learning systems. Humans can readily acquire arbitrary pairwise associations, whereas learning and retaining longer temporal structures is subject to pronounced capacity limits and interference effects.

Requirement 7 (Temporal adaptability). *The model must be able to learn sequences with varying cycle lengths and adapt when the temporal scale changes.*

$$\exists \psi_1, \psi_2 \in \Psi : 0 < |\psi_1| < |\psi_2| \wedge (A_0)_{\psi_1} \triangleright \psi_2.$$

This requirement captures the ability to learn and track temporal structure across multiple timescales. Systems restricted to rigid pattern matching at a single, fixed periodicity are therefore excluded, as adaptive learning requires sensitivity to recurring structure at different temporal scales.

Requirement 8 (Content sensitivity). *Adaptation time is input-dependent.*

$$\exists \psi_1, \psi_2 \in \Psi : |\psi_1| = |\psi_2| \wedge \tau_{A_0}(\psi_1) \neq \tau_{A_0}(\psi_2).$$

The rate at which a model adapts depends on the structure of the input sequence. Simple or highly regular sequences typically yield rapid convergence, whereas less regular inputs require longer exposure before the model can reliably capture the underlying structure.

Requirement 9 (Context sensitivity). *Adaptation time is model-dependent.*

$$\exists \psi \in \Psi, \exists A \neq B : \tau_A(\psi) \neq \tau_B(\psi).$$

A model's internal configuration reflects the cumulative influence of past inputs and thereby determines the context in which new information is learned. When subsequent inputs are consistent with the structure established through prior learning, adaptation may proceed rapidly. Conversely, when new inputs conflict with this learned context, additional iterations may be required before accurate prediction becomes possible.

Requirement 10 (Denoising). *An informed model consistently outperforms the best constant baseline at predicting corrupted inputs.*

Let $\phi = (\mathbf{x}_1, \dots, \mathbf{x}_k) \in \Phi$ be drawn from the underlying stochastic generative process, and let $\phi' = (\mathbf{x}'_1, \mathbf{x}_2, \dots, \mathbf{x}_k)$ be obtained from ϕ by corrupting the first input. Let n satisfy $n \gg |\phi|$, and let $\mathbf{x}_1^* \in X$ satisfy

$$A_0^{\phi^n \phi'} \rightsquigarrow \mathbf{x}_1^*.$$

Then the model's expected match score on the clean input exceeds that of both constant predictors:

$$\mathbb{E}[S((\mathbf{x}_1^*), (\mathbf{x}_1))] > \max\left\{\mathbb{E}[S((\mathbf{0}), (\mathbf{x}_1))], \mathbb{E}[S((\mathbf{1}), (\mathbf{x}_1))]\right\},$$

where $\mathbf{0}, \mathbf{1} \in X$ denote the all-zero and all-one inputs, and the expectation is taken with respect to the sequence generator and the corruption process.

A model is required to recover from a single corrupted input after observing the remaining uncorrupted elements of a previously encountered sequence. When re-exposed to a familiar stimulus, such a model must consistently outperform the best constant baseline when predicting the corrupted input.

Requirement 11 (Real-time liveness). *Each model update completes within a fixed wall-clock time bound, independent of the input history.*

$$\exists t_{\max} > 0 : \Delta t(A, \mathbf{x}) \leq t_{\max} \quad \forall A \in \mathcal{M}, \forall \mathbf{x} \in X,$$

where $\Delta t(A, \mathbf{x})$ denotes the wall-clock time required to perform the atomic transition $A \xrightarrow{\mathbf{x}} A'$.

A model must complete each atomic transition within a bounded amount of time to remain suitable for real-time interaction. Biological brains satisfy this requirement in large part through massive parallelism: configuration transitions and signal emissions occur concurrently across large neuronal populations, with neural signal propagation times bounded under normal operating conditions. As a result, cognitive processing does not exhibit unbounded slowing as a function of accumulated experience or instantaneous sensory load.

A. Immediate consequences of the requirements

Two immediate consequences follow from the preceding requirements. First, any system satisfying Req. 3 undergoes perpetual change: every input induces a distinct internal configuration, and model trajectories contain no exact recurrences. Second, Req. 6 implies a form of behavioural unobservability, in which distinct internal configurations can yield identical autoregressive outputs.

Corollary 1 (Perpetual change). *Every input induces a continual change in the model's internal configuration.*

$$\forall t \geq 0 : A_{t+1} \neq A_t$$

Proof. By Requirement 3, $A_t \neq A_s$ for all $t \neq s$. Setting $s = t + 1$ yields $A_{t+1} \neq A_t$ for all t . $\square$

Corollary 2 (Unobservability). *Distinct model configurations may be observationally indistinguishable under autoregressive generation:*

$$\exists \phi \in \Phi \exists A \neq B : A \Rightarrow \phi \wedge B \Rightarrow \phi.$$

Proof. Choose $\phi \in \Phi \cap X^2$. By Requirement 6(b), every reachable configuration learns ϕ .

Let $A \neq B$ be two reachable configurations with distinct, non-ancestral input histories. By Requirement 3, each input leaves a permanent internal trace; hence these histories cannot be erased or merged by subsequent updates. Therefore, after learning ϕ , the resulting post-learning configurations remain distinct.

Since both configurations learn ϕ , there exist post-learning configurations A_ϕ and B_ϕ such that

$$A_\phi \Rightarrow \phi, \quad B_\phi \Rightarrow \phi, \quad A_\phi \neq B_\phi.$$

$\square$

B. Unstable generalisation

What differentiates AGI from AI is the ability to generalise. Many definitions and metrics exist; a behavioural one suited to the signal-prediction domain is the ability to predict previously unseen input patterns. We cannot, however, operationalise this as an AGITB requirement through a transparent and reusable test, because we have no way to generate a valid generalisation target.

First, a prototype model cannot be expected to generalise to every kind of sequence. Some sequence it could generalise surely exists — but the test has no way of knowing which one, and so may demand generalisation the model was never capable of. Choosing the target externally, therefore, risks handing the model a problem outside its reach, and failure would say nothing about the capability under test.

Second, letting another model of the same kind generate the target fares no better. Here, a fresh model observes a prefix produced by a generator and must predict its continuation, on the assumption that a model should always be able to learn what its own kind can produce. But a finite prefix does not determine the rule that produced it: many rules — all within the fresh model's reach — agree on the prefix yet differ on what follows. The fresh model cannot be *guaranteed* to recover the generator's rule; it may settle on one that fits the prefix but diverges beyond it, and where they diverge the prediction fails — not because the model is too weak, but because the prefix never said enough to fix the rule. Nor does lengthening the prefix repair this: no fixed length suffices, since a generator can always embed a distinction that surfaces only past that length — and demanding ever-longer prefixes itself presumes abilities a simple prototype may lack.

So neither route yields a target the test can fairly demand: an external sequence may lie outside the model's reach, and a self-generated one underdetermines its own generator. This is not a limitation of present constructions but of the setting itself: behaviour underdetermines the generator, so no behavioural test can require its recovery. Generalisation remains the defining capability of AGI, but it is not one

that AGITB can explicitly demand through a transparent and reusable test.

VII. METHODOLOGY

The proposed requirements are operationalised as concrete evaluation procedures, with each requirement evaluated by an empirical test whose failure indicates non-compliance. The objective is not to demonstrate intelligence, but to expose violations under stringent conditions, thereby probing the robustness of candidate systems.

Taken together, the requirement-specific tests form a practical testbed, the Artificial General Intelligence Testbed (AGITB), that cannot, and is not intended to, prove that a model satisfies all twelve requirements. Rather, its strength lies in identifying violations of individual requirements and in indicating which constraints are breached. A reference C++ implementation of AGITB is freely available under the GPL license at <https://github.com/matejsprogar/agitb>.

A. Inherent limitations

AGITB confronts three principal challenges. First, several requirements quantify over effectively unbounded sets of cases and cannot be exhaustively verified through finite testing. Second, individual requirements cannot always be captured by a single, easily interpretable test. Third, external performance metrics are unsuitable, as they cannot reliably distinguish generalisable adaptive behaviour from task-specific optimisation.

We address the first challenge by approximating unbounded domains with finite, tractable collections of test cases that provide empirical support for a given claim. The second is addressed through a synergistic test design in which multiple simple tests are mutually informative and reinforce one another. In such settings, failure to satisfy a requirement may not be revealed by a test in isolation, but instead emerges when constraints imposed by other tests become mutually incompatible. The third challenge is addressed by constructing controlled scenarios involving independent system instances and assessing behaviour through systematic comparison, with success defined in terms of configuration or behavioural distinctions rather than external task scores.

B. Test design

In the reference implementation, a compromise approximation for unlimited iterations is set to 5,000. While finite, this value was chosen to balance computational feasibility with stringency and is used consistently across tests. To minimise the influence of stochastic effects, most tests are repeated 5,000 times, and the system under evaluation must succeed on each trial. The resulting procedure functions as an extreme form of stress testing, ensuring that passing results reflect systematic robustness rather than chance.

The tests are not designed as conventional significance tests, nor to maximise statistical power; instead, they prioritise reliability by admitting virtually no noise or marginal effects. Modest or inconsistent improvements are treated as failures.

Finally, AGITB adopts an all-or-nothing evaluation criterion: a system under evaluation must satisfy all requirement-specific tests to pass the benchmark. While individual tests may be solvable in isolation, the tests are designed such that simultaneous satisfaction of all requirements imposes substantially stronger constraints.

C. Search space

To prevent models from relying on brute-force memorisation, the problem space must be sufficiently large to exceed the capacity of any model operating under realistic computational constraints on both time and memory. In AGITB, tasks typically require predicting a temporal sequence of seven inputs ($N = 7$), each consisting of ten bits ($L = 10$). The resulting search space is $\mathcal{S} = (\{0, 1\}^L)^N$, which has cardinality $|\mathcal{S}| = 2^{70}$, representing all possible binary input sequences of that length.

The *absolute refractory period* prohibits any neuron (bit) from firing in consecutive time steps. This requirement substantially lowers the number of admissible sequences, yielding a subset $\mathcal{S}' \subset \mathcal{S}$. There are $|\mathcal{S}'| = (\mathcal{F}_{N+2})^L = 34^{10} \approx 2^{51}$ distinct seven-step temporal sequences of ten bits under the condition that a 1 never carries over to the next time step, where $\mathcal{F}_i$ denotes the i -th Fibonacci number with $\mathcal{F}_0 = 0$.

In some cases, AGITB further constrains $\mathcal{S}'$ by requiring sequences to be cyclic, such that the first input also satisfies the absolute refractory constraint with respect to the final input in the sequence. The resulting set of cyclic temporal sequences respecting the refractory constraint, denoted $\mathcal{S}'' \subset \mathcal{S}'$, has cardinality $|\mathcal{S}''| = (\mathcal{L}_N)^L = 29^{10} \approx 2^{49}$, where $\mathcal{L}_i$ denotes the i -th Lucas number with $\mathcal{L}_0 = 2$.

The choice of seven-step sequences with ten-bit inputs is sufficient to detect non-AGI behaviour while maintaining computational efficiency. Increasing these default values could exceed the capabilities of the first-generation AGI under evaluation, potentially leading to false negatives and substantially increasing runtime. The current configuration, therefore, ensures that each test remains both computationally feasible and diagnostically informative.

D. Potential Benchmark Exploitation

Because AGITB consists of intentionally simple and transparent tests, it is conceivable that a model could be engineered to optimize for the benchmark rather than for the underlying learning capabilities it is intended to measure. For example, task-specific algorithms could be selectively activated after inferring the currently executed test from observable properties of the evaluation process, such as the number of instantiated models or the sequence of invoked methods. Other potential avenues include exploiting weaknesses in the benchmark's randomisation or exploiting properties of the comparison procedure between model instances. Such strategies optimize performance with respect to the benchmark implementation rather than the underlying capabilities being assessed.

In principle, the benchmark could be hardened against these forms of exploitation by introducing additional randomization,

obfuscation, or verification mechanisms. However, these measures would inevitably increase implementation complexity while reducing transparency, interpretability, and reproducibility. Since AGITB is intended not only as an evaluation protocol but also as a practical development tool, the reference implementation deliberately favours simplicity and readability over resistance to benchmark-specific optimizations.

A more fundamental limitation arises from the need to approximate inherently unbounded evaluation criteria using finite computational resources. Several AGITB requirements would ideally be evaluated over an unbounded number of test cases, which is computationally infeasible. Instead, the benchmark performs a fixed number of iterations intended to approximate an otherwise indefinite evaluation process. In the reference implementation, this limit is controlled by the parameter `SimulatedInfinity`, which is currently set to 5,000.

Although this value is necessarily finite, our experiments indicate that, in combination with the current benchmark configuration (temporal patterns consisting of seven 10-bit input signals), it is sufficient to detect violations of the benchmark requirements while maintaining computational efficiency suitable for rapid iterative experimentation.

The AGITB reference implementation is therefore intentionally designed to be transparent, computationally efficient, and straightforward to inspect. Increasing the complexity of the benchmark solely to discourage hypothetical implementation-specific exploits would provide limited practical benefit while diminishing its value as an interpretable and reproducible evaluation framework.

E. Expected behaviour of representative system classes

To contextualise the benchmark, we consider how representative classes of systems would be expected to behave under AGITB. These observations serve only as qualitative reference points and do not constitute formal verification.

Human cognition provides a useful conceptual baseline. Direct verification of AGITB requirements in humans is not possible because internal cortical configurations cannot be inspected or compared computationally. Nevertheless, many requirements correspond to well-known cognitive abilities, including temporal pattern recognition, denoising of corrupted inputs, and knowledge saturation. One notable exception is the uninformed-start requirement, which adult humans do not satisfy because of accumulated prior knowledge; this condition may plausibly hold only at the start of cortical development.

Classical symbolic programs are generally incompatible with the uninformed start requirement because their rules and representations encode prior assumptions about the meaning of processed signals. In such systems, program structure effectively constitutes external knowledge supplied by the designer. Although individual AGITB tests may be solvable through specialised procedures, the presence of embedded prior knowledge violates the requirement that models derive structure solely from experience with grounded binary inputs.

Artificial neural networks exhibit a similar distinction between pre-informed and uninformed systems. Modern deep

learning architectures typically rely on extensive pretraining, during which network parameters are shaped by exposure to structured datasets. Such pretraining conflicts with the requirement for an uninformed start. In addition, standard ANN training procedures rely on externally supplied optimisation processes rather than autonomous adaptation driven exclusively by incoming signals.

Large language models represent a particular instance of pretrained neural systems and therefore fail the first requirement by construction. Moreover, transformer-based architectures necessarily employ fixed-length context windows, which may prevent preservation of an uninterrupted trace of past inputs when earlier tokens are truncated or compressed. These architectural characteristics place practical limits on their ability to satisfy certain AGITB requirements.

These observations illustrate that while individual AGITB tests may be solvable by specialised systems, satisfying all requirements simultaneously imposes a substantially stronger constraint on candidate architectures.

F. Empirical attempts to satisfy the testbed

To probe whether existing systems might reject the proposed hypothesis, we attempted to construct counterexamples in two ways.

First, the reference implementation of AGITB was released publicly as free software to encourage independent attempts at fulfilling the requirements. The repository was made available, along with a preprint describing the testbed, in April 2025. Although the repository has been cloned several hundred times, no independently verified solution satisfying all tests has been reported.

Second, we evaluated whether contemporary generative AI systems could synthesise a program that could pass the criteria. Several state-of-the-art large language models, including ChatGPT, Claude, and Gemini, were prompted to generate candidate implementations. Although these systems produced programs that they claimed would satisfy the testbed, none of the generated solutions succeeded when executed under the AGITB evaluation procedure. Multiple prompting strategies were explored in this process.

VIII. CONCLUSION

Unlike conventional approaches to defining intelligence through high-level behaviours such as question answering or language translation, this work focuses on low-level, biologically grounded computational properties. The proposed requirements are tightly interdependent: no single requirement is decisive in isolation, yet their complementarity permits each to remain atomic in scope while jointly demanding the adaptive learning capabilities expected of an AGI.

The proposed requirements are not intended as a sufficient criterion for artificial general intelligence, nor do they characterise intelligence in any comprehensive sense. A system that satisfies them does not thereby exhibit higher-level capacities such as reasoning, abstraction, or natural language competence. Instead, the requirements target a set of low-level capabilities that may serve as necessary precursors to more

general forms of intelligence, without themselves guaranteeing their emergence.

The central claim advanced here cannot be verified until a precise, universally accepted definition of intelligence is available, a prospect that is unlikely. It is, however, empirically falsifiable: the existence of a narrow system that satisfies all requirements would invalidate the claim that these properties are necessary for general intelligence.

To date, no artificial system known to the authors has satisfied all requirements simultaneously, despite each being well-defined and individually attainable. In this sense, the proposed requirements delineate a boundary that contemporary systems do not cross and may represent a pathway toward more general forms of intelligence.

CODE AVAILABILITY

The reference implementation of the proposed testbed is freely available at <https://github.com/matejsprogar/agitb>.

FUNDING

The author acknowledges the financial support from the Slovenian Research Agency (research core funding No. P2-0057).

REFERENCES

- [1] A. M. Turing, "Computing machinery and intelligence," *Mind*, vol. 59, no. 236, pp. 433–460, 1950. [Online]. Available: <http://www.jstor.org/stable/2251299>
- [2] L. Guo, H. Chen, Y. Chen, Y. Bi, and S. Yu, "A neural network for modeling human concept formation, understanding and communication," *Nature Computational Science*, 2026, to be published, doi:10.1038/s43588-026-00956-4.
- [3] S. Bringsjord and N. S. Govindarajulu, "Artificial Intelligence," in *The Stanford Encyclopedia of Philosophy*, Spring 2026 ed., E. N. Zalta and U. Nodelman, Eds. Metaphysics Research Lab, Stanford University, 2026. [Online]. Available: <https://plato.stanford.edu/archives/spr2026/entries/artificial-intelligence/>
- [4] N. Maslej, L. Fattorini, R. Perrault, Y. Gil, V. Parli, N. Kariuki, E. Capstick, A. Reuel, E. Brynjolfsson, J. Etchemendy, K. Ligett, T. Lyons, J. Manyika, J. C. Nibbles, Y. Shoham, R. Wald, T. Walsh, A. Hamrah, L. Santarlasci, J. B. Lotufo, A. Rome, A. Shi, and S. Oak, "The ai index 2025 annual report," 2025, arXiv:2504.07139.
- [5] G. Marcus, "Deep learning: A critical appraisal," 2018, arXiv:1801.00631.
- [6] M. Mitchell, "Why AI chatbots lie to us," *Science*, vol. 389, no. 6758, p. eaea3922, 2025, doi:10.1126/science.eaea3922.
- [7] P. Shojaei, I. Mirzadeh, K. Alizadeh, M. Horton, S. Bengio, and M. Farajtabar, "The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity," *SuperIntelligence - Robotics - Safety & Alignment*, vol. 2, no. 6, Sep. 2025, doi:10.70777/si.v2i6.15919.
- [8] E. M. Bender and A. Koller, "Climbing towards NLU: On meaning, form, and understanding in the age of data," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 5185–5198, doi:10.18653/v1/2020.acl-main.463.
- [9] Y. Bisk, A. Holtzman, J. Thomason, J. Andreas, Y. Bengio, J. Chai, M. Lapata, A. Lazaridou, J. May, A. Nisnevich, N. Pinto, and J. Turian, "Experience grounds language," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, B. Webber, T. Cohn, Y. He, and Y. Liu, Eds. Online: Association for Computational Linguistics, Nov. 2020, pp. 8718–8735, doi:10.18653/v1/2020.emnlp-main.703.
- [10] A. Srivastava, A. Rastogi, A. Rao, A. A. M. Shueb, A. Abid, A. Fisch, A. R. Brown, A. Santoro, A. Gupta, A. Garriga-Alonso, A. Kluska, A. Lewkowycz, A. Agarwal, A. Power, A. Ray, A. Warstadt, A. W. Kocurek, A. Safaya, A. Tazarv, A. Xiang, A. Parrish, A. Nie, A. Hussain, A. Askell, A. Dsouza, A. Slone, A. Rahane, A. S. Iyer, A. Andreassen, A. Madotto, A. Santilli, A. Stuhlmüller, A. Dai, A. La, A. Lampinen, A. Zou, A. Jiang, A. Chen, A. Vuong, A. Gupta, A. Gottardi, A. Norelli, A. Venkatesh, A. Gholamidavoodi, A. Tabassum, A. Menezes, A. Kirubarajan, A. Mullokandov, A. Sabharwal, A. Herrick, A. Efrat, A. Erdem, A. Karakas, B. R. Roberts, B. S. Loe, B. Zoph, B. Bojanowski, B. Özyurt, B. Hedayatnia, B. Neyshabur, B. Inden, B. Stein, B. Ekmekci, B. Y. Lin, B. Howald, B. Orinon, C. Diao, C. Dour, C. Stinson, C. Argueta, C. F. Ramirez, C. Singh, C. Rathkopf, C. Meng, C. Baral, C. Wu, C. Callison-Burch, C. Waites, C. Voigt, C. D. Manning, C. Potts, C. Ramirez, C. E. Rivera, C. Siro, C. Raffel, C. Ashcraft, C. Garbacea, D. Sileo, D. Garrette, D. Hendrycks, D. Kilman, D. Roth, D. Freeman, D. Khashabi, D. Levy, D. M. González, D. Perszyk, D. Hernandez, D. Chen, D. Ippolito, D. Gilboa, D. Rodola, D. Drakard, D. Jurgens, D. Datta, D. Ganguli, D. Emelin, D. Kleyko, D. Yuret, D. Chen, D. Tam, D. Hupkes, D. Misra, D. Buzan, D. C. Mollo, D. Yang, D.-H. Lee, D. Schrader, E. Shutova, E. D. Cubuk, E. Segal, E. Hagerman, E. Barnes, E. Donoway, E. Pavlick, E. Rodola, E. Lam, E. Chu, E. Tang, E. Erdem, E. Chang, E. A. Chi, E. Dyer, E. Jerzak, E. Kim, E. E. Manyasi, E. Zheltonozhskii, F. Xia, F. Siar, F. Martínez-Plumed, F. Happé, F. Chollet, F. Rong, G. Mishra, G. I. Winata, G. de Melo, G. Kruszewski, G. Parascandolo, G. Mariani, G. Wang, G. Jaimovitch-López, G. Betz, G. Gur-Ari, H. Galijasevic, H. Kim, H. Rashkin, H. Hajishirzi, H. Mehta, H. Bogar, H. Shevlin, H. Schütze, H. Yakura, H. Zhang, H. M. Wong, I. Ng, I. Noble, J. Jumelet, J. Geissinger, J. Kernion, J. Hilton, J. Lee, J. F. Fisac, J. B. Simon, J. Koppel, J. Zheng, J. Zou, J. Kocoń, J. Thompson, J. Wingfield, J. Kaplan, J. Radom, J. Sohl-Dickstein, J. Phang, J. Wei, J. Yosinski, J. Novikova, J. Bosscher, J. Marsh, J. Kim, J. Taal, J. Engel, J. Alabi, J. Xu, J. Song, J. Tang, J. Waweru, J. Burden, J. Miller, J. U. Balis, J. Batchelder, J. Berant, J. Frohberg, J. Rozen, J. Hernandez-Orallo, J. Boudeman, J. Guerr, J. Jones, J. B. Tenenbaum, J. S. Rule, J. Chua, K. Kanclerz, K. Livescu, K. Krauth, K. Gopalakrishnan, K. Ignatyeva, K. Markert, K. D. Dhole, K. Gimpel, K. Omondi, K. Mathewson, K. Chiafullo, K. Shkaruta, K. Shridhar, K. McDonell, K. Richardson, L. Reynolds, L. Gao, L. Zhang, L. Dugan, L. Qin, L. Contreras-Ochando, L.-P. Morency, L. Moschella, L. Lam, L. Noble, L. Schmidt, L. He, L. O. Colón, L. Metz, L. K. Şenel, M. Bosma, M. Sap, M. ter Hoeve, M. Farooqi, M. Faruqi, M. Mazeika, M. Baturan, M. Marelli, M. Maru, M. J. R. Quintana, M. Tolkiehn, M. Giulianelli, M. Lewis, M. Potthast, M. L. Leavitt, M. Hagen, M. Schubert, M. O. Baitemirova, M. Arnaud, M. McElrath, M. A. Yee, M. Cohen, M. Gu, M. Ivanitskiy, M. Starritt, M. Strube, M. Swędrowski, M. Bevilacqua, M. Yasunaga, M. Kale, M. Cain, M. Xu, M. Suzgun, M. Walker, M. Tiwari, M. Bansal, M. Aminnaseri, M. Geva, M. Gheini, M. V. T. N. Peng, N. A. Chi, N. Lee, N. G.-A. Krakover, N. Cameron, N. Roberts, N. Doiron, N. Martinez, N. Nangia, N. Deckers, N. Muennighoff, N. S. Keskar, N. S. Iyer, N. Constant, N. Fiedel, N. Wen, O. Zhang, O. Agha, O. Elbaghdadi, O. Levy, O. Evans, P. A. M. Casares, P. Doshi, P. Fung, P. P. Liang, P. Vicol, P. Alipoormolabashi, P. Liao, P. Liang, P. Chang, P. Eckersley, P. M. Htut, P. Hwang, P. Miłkowski, P. Patil, P. Pezeshkpour, P. Oli, Q. Mei, Q. Lyu, Q. Chen, R. Banjade, R. E. Rudolph, R. Gabriel, R. Habacker, R. Risco, R. Millière, R. Garg, R. Barnes, R. A. Saurous, R. Arakawa, R. Raymaekers, R. Frank, R. Sikand, R. Novak, R. Sitelew, R. LeBras, R. Liu, R. Jacobs, R. Zhang, R. Salakhutdinov, R. Chi, R. Lee, R. Stovall, R. Teehan, R. Yang, S. Singh, S. M. Mohammad, S. Anand, S. Dillavou, S. Shleifer, S. Wiseman, S. Gruetter, S. R. Bowman, S. S. Schoenholz, S. Han, S. Kwatra, S. A. Rous, S. Ghazarian, S. Ghosh, S. Casey, S. Bischoff, S. Gehrmann, S. Schuster, S. Sadeghi, S. Hamdan, S. Zhou, S. Srivastava, S. Shi, S. Singh, S. Asaadi, S. S. Gu, S. Pachchigar, S. Toshniwal, S. Upadhyay, Shyamolima, Debnath, S. Shakeri, S. Thormeyer, S. Melzi, S. Reddy, S. P. Makini, S.-H. Lee, S. Torene, S. Hatwar, S. Dehaene, S. Divic, S. Ermon, S. Biderman, S. Lin, S. Prasad, S. T. Piantadosi, S. M. Shieber, S. Mishnerghi, S. Kiritchenko, S. Mishra, T. Linzen, T. Schuster, T. Li, T. Yu, T. Ali, T. Hashimoto, T.-L. Wu, T. Desbordes, T. Rothschild, T. Phan, T. Wang, T. Nkinyili, T. Schick, T. Kornev, T. Tunduny, T. Gerstenberg, T. Chang, T. Neeraj, T. Khot, T. Shultz, U. Shaham, V. Misra, V. Demberg, V. Nyamai, V. Raunak, V. Ramasesh, V. U. Prabhu, V. Padmakumar, V. Srikumar, W. Fedus, W. Saunders, W. Zhang, W. Vossen, X. Ren, X. Tong, X. Zhao, X. Wu, X. Shen, Y. Yaghoobzadeh, Y. Lakretz, Y. Song, Y. Bahri, Y. Choi, Y. Yang, Y. Hao, Y. Chen, Y. Belinkov, Y. Hou, Y. Hou, Y. Bai, Z. Seid, Z. Zhao, Z. Wang, Z. J. Wang, Z. Wang,

and Z. Wu, “Beyond the imitation game: Quantifying and extrapolating the capabilities of language models,” 2023, arXiv:2206.04615.

- [11] S. Harnad, “The symbol grounding problem,” *Physica D: Nonlinear Phenomena*, vol. 42, no. 1–3, pp. 335–346, 1990, doi:10.1016/0167-2789(90)90087-6.
- [12] K. Friston, “The free-energy principle: a unified brain theory?” *Nature Reviews Neuroscience*, vol. 11, no. 2, pp. 127–138, 2010, doi:10.1038/nrn2787.
- [13] P. S. Churchland and T. J. Sejnowski, *The Computational Brain*, 2nd ed. Cambridge, MA: MIT Press, 2016.
- [14] R. P. N. Rao and D. H. Ballard, “Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects,” *Nature Neuroscience*, vol. 2, no. 1, pp. 79–87, 1999, doi:10.1038/4580.
- [15] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi:10.1162/neco.1997.9.8.1735.
- [16] J. Hawkins and S. Blakeslee, *On Intelligence*. New York, NY, USA: Times Books, 2005.
- [17] M. Hutter, *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, 2005, doi:10.1007/b138233.
- [18] S. Legg and M. Hutter, “Universal intelligence: A definition of machine intelligence,” *Minds and Machines*, vol. 17, no. 4, pp. 391–444, 2007, doi:10.1007/s11023-007-9079-x.
- [19] A. Clark, “Whatever next? predictive brains, situated agents, and the future of cognitive science,” *Behavioral and Brain Sciences*, vol. 36, no. 3, pp. 181–204, 2013, doi:10.1017/S0140525X12000477.
- [20] F. Chollet, “On the measure of intelligence,” 2019, arXiv:1911.01547.
- [21] W. Maass, “Networks of spiking neurons: The third generation of neural network models,” *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997, doi:10.1016/S0893-6080(97)00011-7.
- [22] W. Gerstner and W. M. Kistler, *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press, 2002, doi:10.1017/CBO9780511815706.
- [23] R. Gubelmann, “Pragmatic norms are all you need – why the symbol grounding problem does not apply to LLMs,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 11663–11678, doi:10.18653/v1/2024.emnlp-main.651.
- [24] J. R. Searle, “Minds, brains, and programs,” *Behavioral and Brain Sciences*, vol. 3, no. 3, pp. 417–424, 1980, doi:10.1017/S0140525X00005756.
- [25] P. Churchland and P. Churchland, “Could a machine think?” *Scientific American*, vol. 262, no. 1, pp. 32–37, 1990, doi:10.1038/scientificamerican0190-32.
- [26] A. Zador, S. Escola, B. Richards, B. Ölveczky, Y. Bengio, K. Boahen, M. Botvinick, D. Chklovskii, A. Churchland, C. Clopath, J. DiCarlo, S. Ganguli, J. Hawkins, K. Körding, A. Koulakov, Y. LeCun, T. Lillicrap, A. Marblestone, B. Olshausen, A. Pouget, C. Savin, T. Sejnowski, E. Simoncelli, S. Solla, D. Sussillo, A. S. Tolia, and D. Tsao, “Catalyzing next-generation artificial intelligence through NeuroAI,” *Nature Communications*, vol. 14, no. 1, p. 1597, 2023, doi:10.1038/s41467-023-37180-x.
- [27] M. Šprogar, “A ladder to human-comparable intelligence: an empirical metric,” *Journal of Experimental & Theoretical Artificial Intelligence*, vol. 30, no. 6, pp. 1037–1050, 2018, doi:10.1080/0952813X.2018.1509897.
- [28] A.-L. Marais and N. Roche-Labarbe, “Predictive coding and attention in developmental cognitive neuroscience and perspectives for neurodevelopmental disorders,” *Developmental Cognitive Neuroscience*, vol. 72, p. 101519, 2025, doi:10.1016/j.dcn.2025.101519.
- [29] A. Darriba, Y.-F. Hsu, S. V. Ommen, and F. Waszak, “Intention-based and sensory-based predictions,” *Scientific Reports*, vol. 11, p. 19899, 2021, doi:10.1038/s41598-021-99445-z.
- [30] S. Cave, “There’s no such thing as free will,” 2016. [Online]. Available: <https://www.theatlantic.com/magazine/archive/2016/06/theres-no-such-thing-as-free-will/480750/>



Matej Šprogar is an Assistant Professor at the University of Maribor, Faculty of Electrical Engineering and Computer Science, Maribor, Slovenia. He received the Ph.D. degree in computer science from the University of Maribor, where his doctoral research focused on autonomous evolutionary construction of decision trees. His research interests include artificial general intelligence, machine learning, and evolutionary computation. Contact him at matej.sprogar@um.si.